

NB3 — Time Travel + MERGE Upsert (lightweight)

Maps to slide §3 + deliverable bullet 3.

Spark equivalent: `option("versionAsOf", N) ↔ DeltaTable(path, version=N)` `MERGE INTO ... ↔ dt.merge(source, predicate)`

```
In [1]: import _setup # noqa: F401 -- adds scripts/ to sys.path
import time
import polars as pl
from deltalake import DeltaTable, write_deltalake
from lakehouse import path, reset

table_path = path("scratch", "customers_tt")
reset(table_path)
```

1. Build version history

v0: initial 100K · v1: schema add · v2: MERGE upsert · v3: bad data

```
In [2]: # v0 - initial Load
v0 = pl.DataFrame({
    "customer_id": list(range(100_000)),
    "status":      ["active"] * 100_000,
    "score":       [i % 1000 for i in range(100_000)],
})
write_deltalake(table_path, v0.to_arrow(), mode="overwrite")

# v1 - add `tier` column (schema evolution)
v1 = (pl.from_arrow(DeltaTable(table_path).to_pyarrow_table())
      .with_columns(
          pl.when(pl.col("score") > 800).then(pl.lit("gold")).otherwise(pl.lit("s
      ))
write_deltalake(table_path, v1.to_arrow(), mode="overwrite", schema_mode="overwrite")

# v2 - MERGE upsert 100K (50K updates, 50K inserts)
updates = pl.DataFrame({
    "customer_id": list(range(50_000, 150_000)),
    "status":      ["vip"] * 100_000,
    "score":       [999] * 100_000,
    "tier":        ["platinum"] * 100_000,
})
t0 = time.time()
(DeltaTable(table_path)
 .merge(source=updates.to_arrow(),
        predicate="t.customer_id = s.customer_id",
        source_alias="s", target_alias="t")
```

```

        .when_matched_update_all()
        .when_not_matched_insert_all()
        .execute()
print(f"MERGE 100K rows: {time.time()-t0:.2f}s")

# v3 - simulate bad data
bad = pl.DataFrame({
    "customer_id": list(range(50)),
    "status":      [None] * 50,
    "score":       [-1] * 50,
    "tier":        ["UNKNOWN"] * 50,
}), schema={"customer_id": pl.Int64, "status": pl.Utf8, "score": pl.Int64, "tier": pl.Utf8}
write_deltalake(table_path, bad.to_arrow(), mode="append")

```

MERGE 100K rows: 0.09s

2. history() — audit trail

```

In [3]: for h in DeltaTable(table_path).history():
        print(f"  v{h['version']}>2}  {h['operation']:<25}  metrics={h.get('operationMe

v 3  WRITE                                metrics={'execution_time_ms': 2, 'num_added_file
s': 1, 'num_added_rows': 50, 'num_partitions': 0, 'num_removed_files': 0}
v 2  MERGE                                metrics={'execution_time_ms': 81, 'num_output_row
s': 150000, 'num_source_rows': 100000, 'num_target_files_added': 1, 'num_target_file
s_removed': 1, 'num_target_files_scanned': 1, 'num_target_files_skipped_during_sca
n': 0, 'num_target_rows_copied': 50000, 'num_target_rows_deleted': 0, 'num_target_ro
ws_inserted': 50000, 'num_target_rows_updated': 50000, 'rewrite_time_ms': 9, 'scan_t
ime_ms': 48}
v 1  WRITE                                metrics={'execution_time_ms': 16, 'num_added_file
s': 1, 'num_added_rows': 100000, 'num_partitions': 0, 'num_removed_files': 1}
v 0  WRITE                                metrics={'execution_time_ms': 16, 'num_added_file
s': 1, 'num_added_rows': 100000, 'num_partitions': 0, 'num_removed_files': 0}

```

3. Time-travel queries

```

In [4]: v0_count = DeltaTable(table_path, version=0).to_pyarrow_table().num_rows
v1_cols  = DeltaTable(table_path, version=1).schema().to_arrow().names
print(f"v0 row count: {v0_count}")
print(f"v1 schema:    {v1_cols}")

```

v0 row count: 100000

v1 schema: ['customer_id', 'status', 'score', 'tier']

4. RESTORE bad version (rollback)

`restore(2)` rewinds the *current* state of the table to whatever it was at version 2, recorded as a new version (v4). The old versions stay in history — restore is itself a transaction, fully auditable.

```

In [5]: t0 = time.time()
dt = DeltaTable(table_path)

```

```
dt.restore(2)
print(f"RESTORE → v2: {time.time()-t0:.2f}s    (target < 30s)")

# Verify the bad rows are gone – use delta-rs's native filter pushdown.
# (DuckDB's delta extension as of 1.5.x is stricter about post-RESTORE
# protocol entries than delta-rs writes; routing through delta-rs end-to-end
# avoids the InvalidProtocolError race.)
dt_after = DeltaTable(table_path)
bad_count = dt_after.to_pyarrow_table(filters=[("score", "<", 0)]).num_rows
print(f"Rows with score<0 after restore: {bad_count}    (expected 0)")
```

RESTORE → v2: 0.01s (target < 30s)

Rows with score<0 after restore: 0 (expected 0)

5. history() — final audit trail (now includes the RESTORE)

```
In [6]: final_history = DeltaTable(table_path).history()
for h in final_history:
    print(f"    v{h['version']:>2}    {h['operation']:<25}")
print(f"\nTotal versions: {len(final_history)}    (target ≥ 5)")
```

```
v 4  RESTORE
v 3  WRITE
v 2  MERGE
v 1  WRITE
v 0  WRITE
```

Total versions: 5 (target ≥ 5)



Deliverable check

- ☐ history() shows ≥ 5 versions (incl. RESTORE itself)
- ☐ MERGE 100K finished in < 60s (likely < 1s on lightweight path)
- ☐ RESTORE finished in < 30s and removed bad rows

```
In [7]: ops = [h["operation"] for h in final_history]
checks = {
    "history ≥ 5 versions": len(final_history) >= 5,
    "history includes the RESTORE": any("RESTORE" in o.upper() for o in ops),
    "MERGE recorded in history": any("MERGE" in o.upper() for o in ops),
    "bad rows gone after restore": bad_count == 0,
}
for k, v in checks.items():
    print(f"    [{ 'PASS' if v else 'FAIL' }] {k}")
assert all(checks.values()), "NB3 incomplete – see FAIL rows above"
print("\nNB3 complete.")
```

```
[PASS] history ≥ 5 versions  
[PASS] history includes the RESTORE  
[PASS] MERGE recorded in history  
[PASS] bad rows gone after restore
```

NB3 complete.